BCS-072	Data Compression	L T P 3 0 0	3 Credits

Pre-requisites: Students should have the knowledge of computer graphics

Course Objectives:

CO1	To introduce students to basic applications, concepts, and techniques of Data Compression
CO2	To develop skills for using recent data compression software to solve practical problems in a variety of disciplines.
CO3	To gain experience doing independent study and research
CO4	Student will be able to understand different types of coding

Detailed Syllabus

MODULE-I

Compression Techniques: Loss less compression, Lossy Compression, Measures of prefonnance,Modeling and coding, Mathematical *Preliminaries* for Lossless compression: A briefintroduction to information theory, Models: Physical models,Probability models, Markov models, composite source model, **Coding:** uniquely decodablecodes, Prefix codes.

The Huffman coding algorithm: Minimum variance Huffman codes, Adaptive Huffman coding:Update procedure, Encoding procedure, Decoding procedure. Golomb codes, Rice codes,Tunstall codes, Applications of Hoffman coding: Loss less image compression, Textcompression, Audio Compression.

MODULE-II

Coding a sequence, Generating a binary code, Comparison of Binary and Huffman cding,Applications: Bi- level image compression-The JBIG standard, JBIG2, Image compression.Dictionary Techniques: Introduction, Static Dictionary: Diagram Coding, Adaptive Dictionary.The LZ77 Approach, The LZ78 Approach, Applications: File Compression-UNIX compress.

Image Compression: The Graphics Interchange Format (GIF), Compression over Modems: V.42bits.Predictive Coding: Prediction with Partial match (ppm): The basic algorithm, The ESCAPE SYMBOL, length of context, The Exclusion Principle, The Burrows-Wheeler Transform: Moveto- front coding, CALIC, JPEG-LS, Multi-resolution Approaches, Facsimile Encoding, Dynamic Markoy Compression.

MODULE-III

Distortion criteria, Models, Scalar Ouantization: The Quantization problem, Uniform Quantizer, Adaptive Quantization, Non uniform Quantization.

Advantages of Vector Quantization over Scalar Quantization, The Linde-Buzo-Gray Algorithm, Tree structured Vector Quantizers. Structured Vector Quantizers.

Text Book:

1. Khalid Sayood, Introduction to Data Compression, Morgan Kaufmann Publishers

Course Outcomes: After the completion of the course the student will be able to:

CO1	Program, analyze Hoffman coding: Loss less image compression, Text compression, Audio Compression
CO2	Program and analyze various Image compression and dictionary-based techniques like static Dictionary, Diagram Coding, Adaptive Dictionary
CO3	Understand the statistical basis and performance metrics for lossless compression
CO4	Understand the conceptual basis for commonly used lossless compression techniques, and understand how to use and evaluate several readily available implementations of those techniques